DFT-Validated Evidence of Ferromagnetic Mislabeling Among Dzyaloshinskii–Moriya-Candidate Materials in the Materials Project Database

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Abstract

The Materials Project (MP) is the usual first filter in searches for Dzyaloshinskii–Moriya-interaction (DMI) hosts, yet its workflow starts every calculation from a ferromagnetic (FM) guess and rarely tests antiferromagnetic (AFM) alternatives. We test MP's stored net-moment labels with density-functional theory (DFT) in the population that DMI screening selects: noncentrosymmetric compounds containing a heavy spin–orbit element. Converging symmetry-distinct AFM orderings for 26 candidates, we find a lower-energy AFM or ferrimagnetic state for 21 (margins 0.19–24.68 meV atom^{-1}), confirm 3 as FM, and find 2 degenerate. For two confirmed ferromagnets, $Li_4Fe_2TeWO_{12}$ and Tb_6FeBi_2, we extract exchange and DMI ($D/J \approx 3.5\%$ and 0.7%) from DFT; atomistic spin dynamics with the unmodified exchange places their ordering temperatures at 101 ± 1 K and 139 ± 3 K. MP ordering labels should be treated as a screening prior, not ground truth, for DMI-material discovery.

INTRODUCTION

Materials that host a stable Dzyaloshinskii–Moriya interaction (DMI) [1, 2] are of active interest for chiral spin-texture and skyrmion-based non-volatile memory, because the antisymmetric exchange term the DMI contributes can stabilize spin spirals and skyrmion lattices at zero field [3–5]. Skyrmion lattices of this kind have been observed in bulk chiral magnets [6, 7], at atomic scale in ultrathin films [8], written and deleted individually with a spin-polarized current [9], and stabilized at room temperature in multilayer stacks [10]; the materials landscape and the device prospects that motivate this search are reviewed in refs. [11, 12]. Searching for such materials computationally typically begins with a curated high-throughput database – most commonly the Materials Project (MP) [13] – used to filter tens of thousands of candidates down to a tractable shortlist before committing expensive first-principles or simulation resources. That filtering step depends on the database's recorded magnetic ordering (ferromagnetic, antiferromagnetic, ferrimagnetic, or non-magnetic) being correct, or at least an unbiased estimate.

Automated high-throughput DFT workflows are prone to a specific failure mode here: relaxing a structure from a single initial magnetic-moment guess – almost always ferromag-

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netic – converges to whichever local minimum that guess is closest to, without necessarily finding the true magnetic ground state [14]. Materials Project's own magnetic-ordering methodology paper is explicit that a full energy-mapping analysis, enumerating symmetry-distinct antiferromagnetic configurations and comparing converged energies, is required to determine the ground state reliably [14], whereas entries generated at high throughput are relaxed from a single ferromagnetic initial guess.

Recent independent evidence, obtained through a very different route, points at the same problem. Fahmy [15] trained propagation-vector and magnetic-order classifiers on the MAGNDATA database of experimentally solved magnetic structures [16, 17] and applied them to the Materials Project, finding a systematic disagreement consistent with a ferromagnetic bias affecting of the order of 7,800 MP-labeled compounds. That is a large-N, purely statistical detection: the classifier disagrees with the label, but no first-principles calculation is performed on any individual compound to confirm which side is right.

Here we ask the complementary question at higher fidelity and lower N: within the structural family that DMI-candidate screening pipelines actually select – noncentrosymmetric space groups, a net magnetic moment, and a heavy spin-orbit-coupling (SOC) element – how often is MP's stored FM label the true DFT ground state? We build a seven-source harmonized materials database and a structure-aware magnetic-ordering classifier, use them to filter for DMI-compatible candidates, and run the collinear energy-mapping protocol MP itself prescribes [14] on 26 shortlisted candidates. The result is a per-compound DFT verdict rather than a statistical disagreement: 21 of the 26 net-moment labels (24 FM, 2 ferrimagnetic) are overturned. Twenty-four of the 26 candidates, and all three confirmed ferromagnets, are hypothetical structures generated by MP's substitution workflows, which is itself a property of what DMI screening selects. We then carry two confirmed ferromagnets, $Li_4Fe_2TeWO_{12}$ (mp-754090) and Tb_6FeBi_2 (mp-1079285), from DFT exchange and DMI constants to finite-temperature atomistic spin dynamics with no fitting or rescaling, and fix the ordering temperature of each model by finite-size scaling; for the third, $FeP_6(WO_8)_3$ (mp-850225), we resolve its two exchange shells.

RESULTS

Screening 64,635 records to a DMI-compatible shortlist

Seven public sources – Materials Project [13], JARVIS-DFT [18], AFLOW [19], MAGNDATA [16, 17], 2DMatPedia [20], OQMD [21, 22] and NOMAD [23] – were merged into one schema of 64,635 records (Methods). Where no ground-truth ordering exists, it is predicted by a LightGBM [24] classifier on composition, global-structure and local-magnetic-geometry features, which reaches 67.4% accuracy (68.0% balanced accuracy) on a stratified held-out set of 7,193 records (Supplementary Fig. 1). Its dominant confusion is between FM and ferrimagnetic (FiM), the two net-moment classes that the screening treats identically. Graph neural networks trained and tested on the same split – a crystal graph convolutional network learned from scratch and LightGBM probes of the frozen MACE-MP-0 foundation model – reach 63.3–74.6% accuracy (65.9–71.7% balanced; Supplementary Table 1, Methods). The best, which appends the MACE-MP-0 embedding to the screening features, gains 7 points of accuracy, but in agreement with Materials Project's own labels: on the experimentally labelled MAGNDATA subset every model lies between 49 and 56% balanced accuracy, within the sensitivity of that small set. Retrained with the 26 DFT-validated candidates of this work held out, the models call 7–13 of the 21 that DFT overturns non-ferromagnetic, 1.5–2.8 times their rate on Materials Project ferromagnets. The screening below uses the LightGBM classifier throughout.

The database is then filtered for candidates plausibly compatible with a DMI (Fig. 1a): a net magnetic moment (FM or FiM), a noncentrosymmetric space group (DMI is symmetry-forbidden under an inversion centre [2]), at least one heavy-SOC element (Ta, W, Re, Os, Ir, Pt, Au, Hg, Bi – the classic DMI-enhancing species of the Fert–Levy and Moriya mechanisms [2, 25–27]), and thermodynamic plausibility. Of 64,635 records, 47,614 carry a net-moment ordering, 19,280 of those are noncentrosymmetric, and 1,864 also contain a heavy-SOC element (Materials Project 768, JARVIS 537, AFLOW 392, 2DMatPedia 77, NOMAD 53, OQMD 35, MAGNDATA 2). Dropping every record with a known hull distance above 100 meV atom^{-1} (sources without a hull field are retained) leaves 1,195 (Fig. 1b,c). The heavy-SOC filter shifts the population sharply towards AFLOW, and the hull filter towards the sources with no hull energy to test. After a structure-backfill and machine-learned-

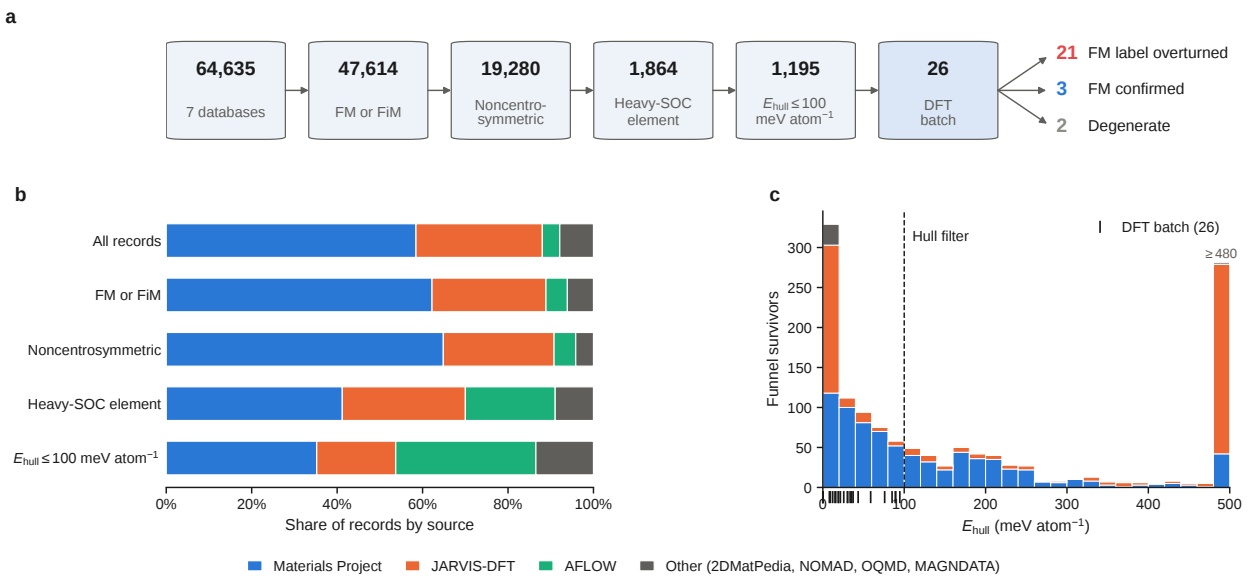


Fig. 1. **Screening a seven-source database for DMI-compatible ferromagnets.** **a**, The screening funnel: net-moment ordering (ground-truth label where present, otherwise the classifier's prediction), noncentrosymmetric space group, at least one heavy-SOC element, and a known hull distance no more than 100 meV atom^{-1} above the hull (records without one are retained), ending in the 26-candidate Materials Project batch tested with DFT and its verdicts. **b**, Share of the records at each stage by source database (legend shared with **c**); "Other" pools 2DMatPedia, NOMAD, OQMD and MAGNDATA. **c**, Hull distance of the 1,340 heavy-SOC survivors that carry one (Materials Project and OQMD as stored, JARVIS-DFT recomputed from its own formation energies; Methods), in 20 meV atom^{-1} bins with values $\geq$ 480 meV atom^{-1}gathered in the last bin; dashed line, the hull filter; ticks below the axis, the 26 DFT candidates.

potential consistency check of the top of each source's ranking (Methods), the resulting shortlist is the pool from which the DFT batch of 26 Materials Project candidates is drawn, every one carrying a confidence-1.0 net-moment label recorded by MP (24 FM, 2 FiM). All 26 survive the hard hull filter, and they spread from rank 1 to rank 663 of the hard-filtered ranking, so the batch is a sample of the funnel's DMI-compatible population rather than of its most stable members (Supplementary Fig. 2).

DFT overturns the stored ordering of 21 of 26 candidates

For each candidate we enumerate FM and symmetry-distinct AFM configurations and relax them with MP's own GGA+U (generalized gradient approximation plus Hubbard U) recipe, audit every calculation for self-consistency and for the intended per-ion spin state, and re-decide every small margin with a same-cell ("tight") FM/AFM pair (Methods). A label is called overturned when an audited non-FM configuration lies more than 0.10 meV atom^{-1} below FM – twice the per-atom electronic-convergence tolerance of the MP protocol – and confirmed or degenerate only when every staged configuration has a valid energy.

Of the 26 candidates, 21 have their FM label overturned by a lower-energy antiferromagnetic or ferrimagnetic configuration (margins 0.19–24.68 meV atom^{-1}), 3 are confirmed as the true FM ground state (1.24–2.23 meV atom^{-1} below the next-lowest configuration), and 2 ($CaLaTaFeO_6$ with AFM 0.10 meV atom^{-1} below FM and $Ba_4Ta_{10}FeO_{30}$ at 0.00 meV atom^{-1}, both from tight pairs) are degenerate rather than evidence for either ordering (Fig. 2a, Table 1). No candidate remains unresolved: the one configuration never completed (the $1\times1\times2$ stacking of mp-682554) belongs to a candidate that another audited AFM cell already places 0.31 meV atom^{-1} below FM. The two shortlisted candidates that MP labels FiM (mp-22972, mp-551086) are counted against the FM alternative only, since a collinear FM/AFM enumeration cannot represent MP's recorded ferrimagnetic configuration directly; both have a collinear configuration below FM (an AFM cell 0.46 meV atom^{-1} below the FM twin of the same cell for mp-22972, a ferrimagnetic Fe/Cr cell 0.80 meV atom^{-1} below FM for mp-551086, both from tight pairs). That antiferromagnetic alignment wins this often is unsurprising on chemical grounds: 23 of the 26 candidates are oxides, in which the magnetic cations are bridged by oxygen – the geometry for which Goodenough–Kanamori–Anderson superexchange between half-filled d shells favours antiparallel alignment [28–30]. A workflow that initializes every relaxation ferromagnetically therefore starts each of these compounds from the chemically less likely state.

The tight pairs changed four verdicts, in both directions. Zn_2FeWO_6 (mp-1047414), degenerate at 0.10 meV atom^{-1} under the MP recipe, is AFM by 0.17–0.19 meV atom^{-1} in both of its same-cell pairs; $Ba_4Ta_{10}FeO_{30}$ (mp-1228547), "confirmed FM" by 1.18 meV atom^{-1} under the recipe, is exactly degenerate (0.001 meV atom^{-1}) once FM and AFM are compared in the same 90-atom cell; $TiFeBiPbO_6$ (mp-1216981), whose enumerated "AFM" cells had

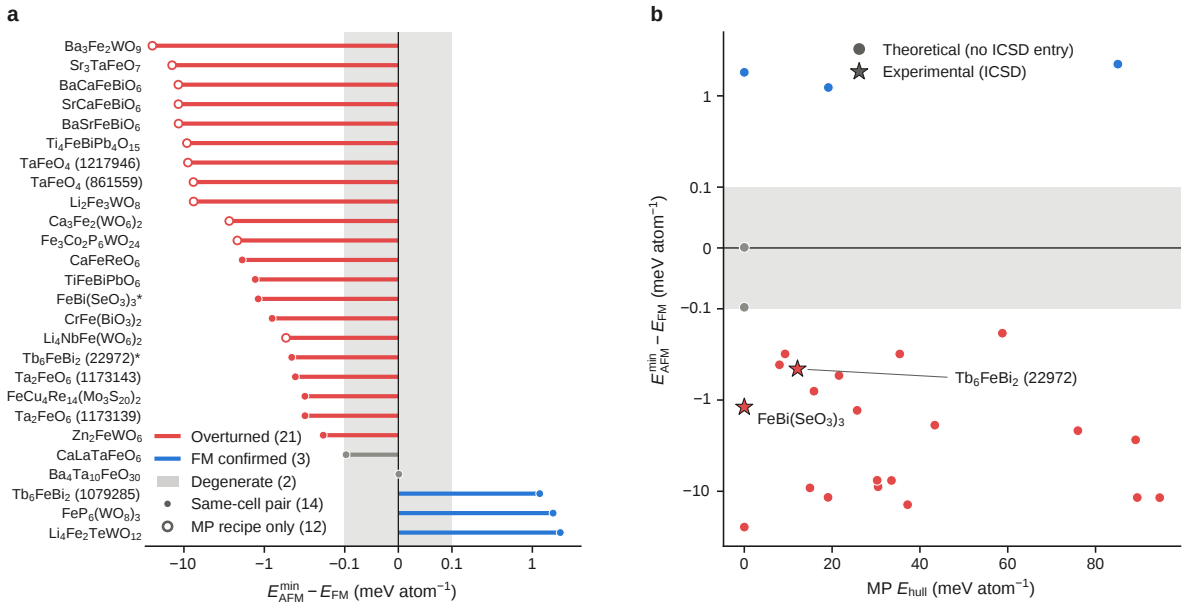


Fig. 2. **DFT verdicts on the ordering labels of 26 Materials Project candidates. a**, Energy of the lowest audited non-FM configuration relative to FM, per atom, for each candidate (symmetric-logarithmic axis, linear within ± 0.1 meV atom^{-1}); negative values overturn MP's FM label. Only calculations that converged and passed the spin-state audit enter, and a same-cell pair replaces the MP-recipe number wherever one exists. The gray band is the ± 0.10 meV atom^{-1} degeneracy window; asterisks mark the two experimentally reported (ICSD) compounds; duplicate formulas carry their MP identifier. The two Tb_6FeBi_2 entries fall on opposite sides: the hypothetical polymorph mp-1079285 is a confirmed ferromagnet, the experimental mp-22972 is overturned. Filled end markers, margins from a same-cell (tight) pair; open, MP-recipe comparison only. **b**, The same margin against MP's stored hull distance. Stars, ICSD compounds; circles, hypothetical structures with no ICSD entry. Every confirmed ferromagnet is hypothetical, and the margin shows no trend with hull distance. $n = 26$ candidates; one point per candidate.

flipped only the d^0 Ti site, is AFM by 0.80–1.30 meV atom^{-1} in all four Fe-only supercells; and Tb_6FeBi_2 (mp-1079285) and $FeP_6(WO_8)_3$ (mp-850225) remain FM, with same-cell margins on the enumerated AFM configurations (3.25 and 3.38 meV atom^{-1}) larger than the recipe numbers (2.32 and 2.09) suggested. In both compounds an ordering computed later for the exchange model sets the final margin: the inter-chain AFM orderings of mp-1079285 lie 1.24–1.25 meV atom^{-1} above FM, and a 68-atom cell of mp-850225 with alternate Fe layers antiparallel along the hexagonal c axis lies 1.81 meV atom^{-1} above FM. For mp-850225,

Table 1. **Summary of DFT ordering verdicts for the 26 candidates.** "Tight" counts the verdicts whose deciding margin comes from a same-cell FM/AFM pair (Methods); the remaining verdicts rest on the MP-recipe comparison, one of them with a margin below 2 meV atom^{-1} (Discussion). The column counts candidates, not same-cell pairs; several candidates carry more than one pair (Methods).

Outcome	Count	Tight
MP FM label overturned (AFM/FiM)	21	9
FM confirmed	3	3
Degenerate (≤ 0.10 meV atom^{-1})	2	2
Total	26	14

"FM" describes the Fe sublattice, which is the only part the enumeration varies: its W^{5+} moments (Methods) align antiparallel to the Fe, so the DFT ground state is strictly a collinear ferrimagnet, $2.0\,\mu_B$ per formula unit against $4.4\,\mu_B$ from Fe alone. Every overturn whose MP-recipe margin was below 2 meV atom^{-1} was re-decided with a tight pair in the cell of its lowest valid non-FM configuration, and every one kept its sign, with margins moving by at most 0.32 meV atom^{-1} (Supplementary Table 2); for $Li_4NbFe(WO_6)_2$ the tight pair (-0.47 meV atom^{-1}) lies just above a second, recipe-only configuration (-0.54 meV atom^{-1}), which therefore sets its margin. Fig. 2a marks which margins come from a same-cell pair.

Among the 24 candidates that resolve clearly, overturned outcomes outnumber confirmed FM outcomes 21 to 3. The batch is drawn from the top of the DMI-filtered shortlist rather than from a random or stratified sample of MP's labelled population, so it establishes the mislabeling rate within this structural family rather than a database-wide estimate; it is plausible, though untested here, that this family is more AFM-prone than the population as a whole. What it does establish is that MP's FM label cannot be taken at face value for candidates drawn from exactly the population that DMI- and skyrmion-material screening pipelines – including the one used here – select by construction.

Confirmed ferromagnets are hypothetical structures

A provenance check against MP's own metadata (`theoretical` flag and Inorganic Crystal Structure Database (ICSD) [31] cross-references, queried 2026-09-20) adds a caveat that applies to the shortlist as a whole rather than to the ordering verdicts. Twenty-four of the 26 candidates are flagged theoretical – structures generated by MP's substitution and prototype workflows, 23 of them with no ICSD entry at all (mp-551086 carries an ICSD identifier but is still flagged theoretical). Only mp-1197024 ($FeBi(SeO_3)_3$, DFT ground state AFM) and mp-22972 (Tb_6FeBi_2, five ICSD entries) are experimentally reported, both overturned, and every candidate whose FM label survives DFT is hypothetical, with MP hull distances of 0–85 meV atom^{-1}(Fig. 2b). The Tb_6FeBi_2 pair makes the point directly: the experimental compound is AFM in DFT, while the confirmed ferromagnet mp-1079285 is a hypothetical polymorph of it. The funnel's symmetry and composition filters therefore preferentially select MP's generated structures, which is unsurprising – noncentrosymmetric cation orderings of otherwise centrosymmetric prototypes are exactly what substitution workflows produce – but it means a confirmed-FM verdict here is a statement about a candidate that may not be synthesizable, and any experimental follow-up should start from the two ICSD-backed entries.

First-principles exchange and DMI of two confirmed ferromagnets

For $Li_4Fe_2TeWO_{12}$ (mp-754090, space group $P1$), confirmed FM by 2.23 meV atom^{-1}, the Fe sublattice is close-packed: each Fe has six neighbours in a triangular layer (5.11–5.16 Å) and six across the layer spacing (5.41–5.43 Å) (Fig. 3a). The same-cell FM/AFM pair gives one effective exchange for all twelve bonds, $J = 2.78$ meV per Fe–Fe bond ($zJ = 12J = 33.4$ meV), with an 8% protocol spread that we carry as its systematic uncertainty, and a constrained noncollinear canting probe resolves one projection of the DMI vector, $D_y = 0.097$ meV per bond, so $D/J \approx 3.5\%$ (Methods). This small ratio is physically plausible: the magnetic ion at the probed bond, Fe, is a light $3d$ species with weak intrinsic SOC, and the tungsten that qualified this candidate for the funnel enters only through the ligand field. The presence of a heavy-SOC element anywhere in the composition does not guarantee a large DMI at the magnetic-ion bond that anchors a spin texture.

Tb_6FeBi_2 (mp-1079285, space group Pm, one Fe per primitive cell) is a different magnetic topology: the Fe atoms form chains along b (Fe–Fe 4.198 Å), with the chain next-nearest neighbours ($\pm 2b$) and six inter-chain neighbours all at 8.40–8.41 Å (Fig. 3b). Four same-cell pairs resolve three exchange shells. The chain pair places the chain-antiparallel state 58.43 meV per cell above FM, $J_b = 14.61$ meV. The two inter-chain pairs are nearly identical, $\Delta E_{2a} = 22.48$ and $\Delta E_{2c} = 22.24$ meV per 18-atom cell (1.25 and 1.24 meV atom^{-1}; Fe ± 1.64–$1.67\,\mu_B$, induced Tb $\leq 0.21\,\mu_B$ in all four calculations), which supports the $J_a \approx J_c$ part of the mean-field assumption and gives $J_\perp = 2.79$ meV – a factor of 5.2 weaker than the chain coupling, but ferromagnetic, so the material is a three-dimensional ferromagnet of coupled FM chains rather than a quasi-one-dimensional one. The $1\times4\times1$ $++--$ pair places that state 84.92 meV per 36-atom cell above FM; subtracting $4J_b$ leaves $8J_{2b} = 26.50$ meV, i.e. $J_{2b} = 3.31$ meV – ferromagnetic and comparable to $J_\perp$, so the $\pm 2b$ bond is not a negligible correction: it carries 13% of $zJ = 2J_b + 2J_{2b} + 6J_\perp = 52.6$ meV and raises the mean-field estimate from $T_c^{\mathrm{MF}} = 178$ K (chain + inter-chain only) to 203 K. By symmetry the chain-bond DMI vector is perpendicular to b, and two pairs of 90° spin spirals fix both of its components, $\mathbf{D}\cdot\hat{n}_1 = -0.074$ and $\mathbf{D}\cdot\hat{n}_2 = -0.063$ meV: $|\mathbf{D}| = 0.098$ meV per chain bond and $|\mathbf{D}|/J_b = 0.67\%$, five times smaller than the D/J of mp-754090, again with Fe as the only magnetic ion and the heavy elements (Tb, Bi) contributing only through the ligand field.

J_{2b} is a difference of two energies computed with different k-meshes, so we checked it with a same-cell pair: the $+-+-$ chain pattern in the $1\times4\times1$ cell, which flips all four chain bonds ($\Delta E = 8J_b$), gives $\Delta E = 100.45$ meV per 36-atom cell, i.e. $J_b = 12.56$ meV, 14% below the chain-pair value. The two pairs differ in cell shape and k-mesh (Methods), so the spread is the cell-shape and k-convergence sensitivity of a metallic Fe chain, and it brackets the exchange model: with the same-cell J_b the residual is $J_{2b} = 4.34$ meV, $zJ = 50.6$ meV and $T_c^{\mathrm{MF}} = 196$ K, against $J_{2b} = 3.31$, $zJ = 52.6$ meV and 203 K with the chain-pair value; $|\mathbf{D}|/J_b$ ranges over 0.67–0.78% (Fig. 3c, open markers). The production model uses the chain-pair parameters.

The third confirmed ferromagnet, $FeP_6(WO_8)_3$ (mp-850225), has two exchange shells, six in-plane 8.77 Å and six out-of-plane 8.91 Å Fe–Fe bonds. A second same-cell pair, with alternate hexagonal Fe layers antiparallel along c, flips every out-of-plane bond and no in-plane bond, so $\Delta E = 12J_2 = 123.2$ meV per 68-atom cell, giving $J_2 = 10.3$ meV and, with $8(J_1 + J_2) = 229.6$ meV from the first pair, $J_1 = 18.4$ meV, both ferromagnetic.

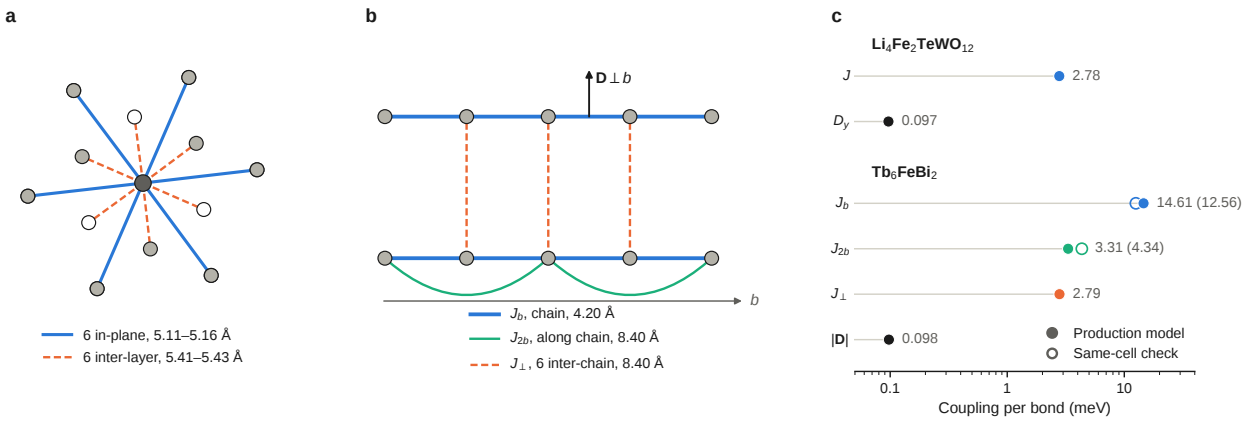


Fig. 3. **Exchange paths and DFT couplings of the two modelled ferromagnets. a**, $Li_4Fe_2TeWO_{12}$ (mp-754090): the twelve Fe neighbours of one Fe (centre), projected on the plane of the six in-plane bonds (solid); dashed, the six inter-layer bonds; filled and open circles lie above and below that plane. One same-cell FM/AFM pair fixes a single J for all twelve bonds; a constrained canting probe resolves the y projection of the DMI. **b**, Tb_6FeBi_2 (mp-1079285): two neighbouring Fe chains along b, drawn to scale, with the chain bond J_b, the next-nearest chain bond J_{2b} (arcs, shown on one chain) and the inter-chain bond $J_\perp$ (dashed; six per Fe, one direction drawn); the Pm mirror forces the chain-bond DMI vector **D** perpendicular to b. **c**, The couplings per bond on a logarithmic axis. Filled markers, the production spin models; open markers, the J_b of the same-cell $+-+-$ check and the J_{2b} it implies, which bracket the Tb_6FeBi_2 model.

The FM members of the two pairs (the same 68 atoms in different supercells) agree to 0.006 meV atom^{-1}. Because the W^{5+} moments are free to follow the Fe, these are effective Fe–Fe couplings that include the W polarization. Its DMI could not be resolved: with one Fe per rhombohedral cell a two-sublattice canting probe sees no DMI at all, and of the spin-spiral pair that would resolve it in the 102-atom hexagonal cell, one chirality never reached self-consistency at the precision the extraction needs (Methods).

Spin models order at 101 K and 139 K by finite-size scaling

The DFT exchange, with no fitting or rescaling, parameterizes classical Heisenberg models of the Fe sublattices that are run with Langevin spin dynamics in LAMMPS-SPIN [32] from the ordered state (Methods). Each model reproduces every DFT energy difference it was built from in a zero-step evaluation before any sweep.

For mp-754090, $|m|$ of the 2048-spin cell decreases monotonically from 0.966 at 10 K through 0.805 at 50 K and 0.659 at 75 K to 0.343 at 100 K, collapses to 0.068 at 125 K, and then sits on the finite-size floor (0.025–0.044 for 150–400 K; the $1/\sqrt{N}$ value for $N = 2048$ is 0.022) (Fig. 4a, Supplementary Table 3). An independent Metropolis Monte Carlo calculation [33] of the same Hamiltonian on the same lattice gives $|m| = 0.967$, 0.912, 0.806, 0.661, 0.37±0.03 and 0.07±0.02 at 10, 25, 50, 75, 100 and 125 K, agreeing with the Langevin sweep to within 0.002 below 100 K and within the seed spread at 100 K, so the curve is the equilibrium $|m|(T)$ of the model and not an artifact of the thermostat. A control sweep that adds a DMI of the measured magnitude changes $|m|$ by at most 0.013 (at 100 K, within the seed spread) and by ≤ 0.005 elsewhere, so a 3.5% antisymmetric term does not move the ordering scale of this model.

Finite-size scaling locates the transition. With the box aspect ratio held fixed at $N = 256$, 864, 2048 and 4000 spins, the fourth-order Binder cumulant [34] $U_4 = 1 - \langle m^4\rangle/3\langle m^2\rangle^2$ of consecutive sizes crosses at 99.5, 101.3 and 102.7 K (Fig. 4b). The crossings drift upward monotonically with system size, the expected signature of corrections to scaling, giving $T_c = 101.2\pm1.3$ K, with the largest pair ($N = 2048\times4000$, least affected by those corrections) at 102.7 K; the susceptibility $\chi = N(\langle m^2\rangle - \langle|m|\rangle^2)/k_BT$ peaks at 105 K independently at all four sizes. The transition of this model is therefore the classical-Heisenberg fcc value, $k_BT_c \approx 3.18\,J$ or ≈ 103 K [35], to within the 5 K grid spacing, while mean field ($T_c^{\mathrm{MF}} = zJ/3k_B = 129$ K) overestimates it by 28%. The 8% protocol spread in J shifts the transition by about 8 K.

For the three-shell mp-1079285 model, $|m|$ falls smoothly from 0.988 at 5 K to 0.672 at 100 K and 0.509 at 125 K, then collapses to 0.128 at 150 K (the one point with a seed-to-seed spread, ±0.012, as expected at the transition of a finite cell) and sits on the finite-size floor of 0.03–0.05 from 200 K (Fig. 4c, Supplementary Table 4). The same-cell parametrization, swept at 10, 100, 125, 150 and 200 K, gives $|m| = 0.975$, 0.668, 0.497, 0.124 and 0.047 against 0.975, 0.672, 0.509, 0.128 and 0.048 for the production model, so the 4% change in zJ does not move the collapse from between 125 and 150 K. The sweep is insensitive to the DMI: $|\mathbf{D}|/J_b = 0.67\%$ produces a canting angle of order $D/2J \approx 0.2^\circ$ per bond, far below the thermal disorder at any temperature on the grid.

The $U_4(T)$ curves of consecutive sizes cross at 140.9, 141.4 and 135.6 K, giving $T_c = 139.3 \pm 2.6$ K, and χ peaks at 145 K for the three smaller cells and 140 K for the largest

(Fig. 4d). This puts the transition at $0.68\,T_c^{\mathrm{MF}}$, just below the isotropic simple-cubic ratio 0.72 (0.79 for fcc [35]), as the exchange anisotropy leads one to expect. The crossings are not monotonic in system size as they are for mp-754090, and the analysis is harder here: the exchange is strongly anisotropic ($J_b/J_\perp = 5.2$, though ferromagnetic on every shell), so intra-chain correlations survive well above the three-dimensional ordering temperature and the U_4 curves separate gradually rather than pivoting about a point, and the fluctuations near T_c are correspondingly larger. The 135–150 K window was therefore run with 10–14 seeds per point, six of them 80 ps; the three crossings then agree to ± 2.6 K. Both spreads over pairwise crossings are lower bounds on the uncertainty of the model T_c: re-deriving the crossings under changes to the sampling window shows that the autocorrelation time of U_4 near T_c is comparable to the run length (Methods), and the DFT coupling uncertainties come on top.

DISCUSSION

Fahmy's classifier [15] detects the Materials Project's ferromagnetic bias statistically, at large N; the present work reaches the same conclusion by the opposite route, a small-N DFT energy-mapping check, in exactly the noncentrosymmetric, heavy-SOC-element, net-moment slice that DMI- and skyrmion-material pipelines target. A 21-of-26 overturn rate in that slice is a direct warning to any workflow, including our own, that uses MP's ordering label as a hard screening criterion rather than a prior.

The consequence runs in both directions. The funnel used here excludes AFM-labeled candidates, on the reasoning that a net moment is required for the texture-hosting mechanisms it targets. If 21 of 26 top-ranked FM-labeled candidates are in fact AFM or ferrimagnetic, downstream compute – structure backfill, potential relaxation, spin dynamics – risks being spent on candidates whose net-moment premise is wrong, while part of the true net-moment landscape may be filtered out before any calculation is spent on it. The corrective is not to trust a classifier over MP either – both are estimates – but to treat any ordering field, database-recorded or predicted, as a prior to be checked with DFT before committing expensive downstream stages. The provenance result adds a second, independent prior: in this family the screening preferentially returns hypothetical structures (24 of 26 here, and all three confirmed ferromagnets), so a verified magnetic ground state still has to be paired

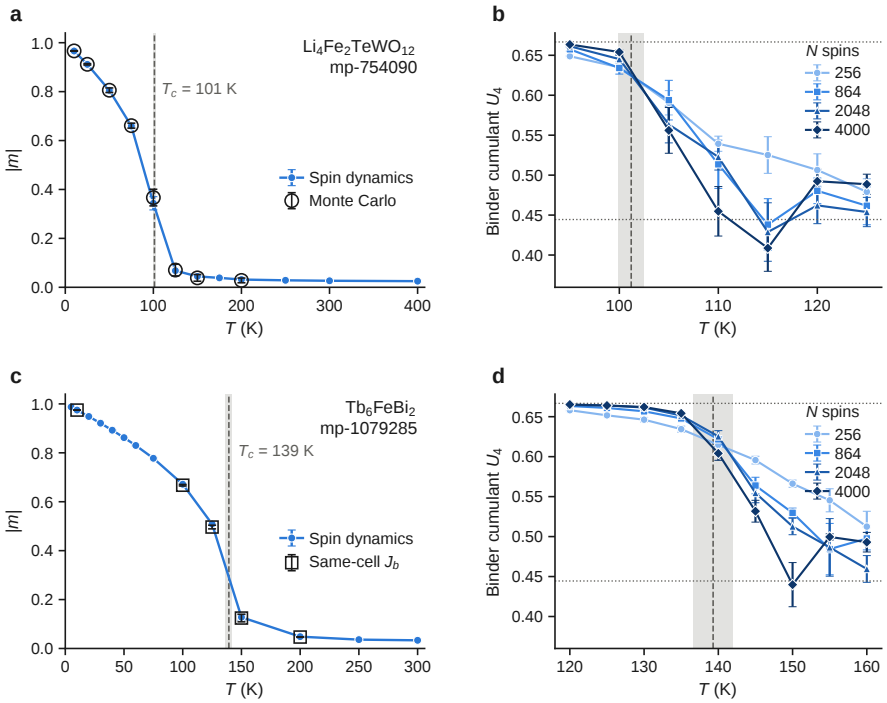


Fig. 4. **Finite-temperature ordering of the two DFT-parameterized spin models. a**, $Li_4Fe_2TeWO_{12}$ (mp-754090): normalized Fe-sublattice magnetization $|m|$ of the 2048-spin cell from 20 ps Langevin dynamics ($\alpha = 1$) started from the ordered state, averaged over the last 10 ps; filled circles, mean ± s.d. over $n = 3$ thermostat seeds per temperature (error bars mostly smaller than the symbols); open circles, independent Metropolis Monte Carlo of the same Hamiltonian on the same lattice (mean ± s.d. over 20 blocks of the 2000 measurement sweeps). **b**, Binder cumulant U_4 at four system sizes (N, box aspect ratio fixed) from 40 ps runs, moments pooled over the last 20 ps of $n = 4$ seeds; error bars, drop-one-seed jackknife. Dotted lines, the ordered (2/3) and disordered (4/9) limits of a three-component order parameter. **c**, **d**, As **a**, **b** for the three-shell Tb_6FeBi_2 (mp-1079285) model; open squares in **c**, the same-cell parametrization ($n = 3$ seeds). In **d**, $n = 10$–14 seeds at 135–150 K (six of them 80 ps) and $n = 4$ elsewhere. In every panel the dashed line and gray band are T_c and its spread, the mean ± s.d. of the three pairwise crossings of consecutive sizes.

with a synthesizability assessment before it is an experimental target.

For two confirmed candidates, the full chain from DFT to finite temperature costs one and five same-cell pairs and two and four constrained noncollinear calculations, respectively, plus a finite-size-scaling campaign, and returns ordering temperatures of the models, 101 ± 1 K

and 139±3 K, with no adjustable parameter. Both DMI-to-exchange ratios are a few percent or less, despite the heavy elements that qualified the compounds for screening, which argues for adding a magnetic-ion-resolved SOC criterion to DMI screening rather than an element-presence one.

Several limitations bound these conclusions. The 26-candidate batch is drawn from the top of one DMI-filtered shortlist, not a random sample of MP's labelled population (1,864 candidates pass the DMI funnel, 1,195 with the hull as a hard filter), so the mislabeling rate is a property of this structural family. The verdict rule is deliberately asymmetric, and the 12 overturns decided on an MP-recipe pair alone carry a protocol uncertainty that the sixteen doubly-computed configurations bound at 1.3 meV atom^{-1}; only one of them has $|\text{margin}| < 2$ meV atom^{-1} ($Li_4NbFe(WO_6)_2$, -0.54), and a tight pair on a second configuration corroborates its sign (-0.47 meV atom^{-1}). The mp-754090 DMI probe resolves one projection of what is, in $P1$ symmetry, eight unconstrained three-component DMI vectors, so its spin model is exchange-only with the DMI bounded by a control; the DMI of mp-850225 is not resolved. The spin models are classical, unit-spin, rigid-lattice models with zero single-ion anisotropy, which was not probed in any compound; the mp-754090 model assigns one J to two geometrically distinct shells, the mp-1079285 model one $J_\perp$ to six inter-chain bonds, and Tb is treated with its $4f$ electrons in the core, so any Tb–Fe exchange is absent. The Binder-cumulant crossings are therefore critical temperatures of the spin models, not of the materials: they inherit an 8% spread in J for mp-754090 and a 14% spread in J_b for mp-1079285 on top of the classical treatment of the moment.

Within the structural family that DMI-candidate screening selects, MP's ferromagnetic label is wrong more often than it is right. High-throughput ordering labels should be used as a screening prior rather than ground truth, particularly in the structural families that a downstream physical mechanism – here, the DMI – selects for.

METHODS

Harmonized multi-source database

Materials Project [13] (37,783 records, via the `mp-api` REST interface [36]), JARVIS-DFT [18] (19,116), AFLOW [19] (2,639), MAGNDATA [16, 17] (1,919 of 2,414 attempted),

2DMatPedia [20] (1,862), OQMD [21, 22] (816), and NOMAD [23] (500) were merged into one schema (`material_id`, `formula`, `elements`, `space_group`, `density`, `volume`, `band_gap`, `ordering`, `total_magnetization`, and, where available, full structure and per-site magnetic-moment vectors), for 64,635 total records. Records retain a source tag rather than being silently deduplicated across sources. MAGNDATA is the only source with experimentally determined (rather than DFT-derived) magnetic structure, and is accordingly the ground-truth source for classifier training and validation. A scraper against the live Bilbao Crystallographic Server index recovered 1,919 of 2,414 entries (79.5%); the failures, mostly transient server errors, are analysed in Supplementary Note 1. Ordering was validated on a known case ($LaMnO_3$, correctly recovered as AFM, space group *Pnma*).

Structure-aware magnetic-ordering classifier

A LightGBM [24] gradient-boosted-decision-tree classifier predicts ordering (FM, AFM, FiM or nonmagnetic, NM) from composition- and structure-derived features, in the spirit of general-purpose composition-based property models [37, 38] and of earlier machine-learning treatments of magnetic order and ordering temperature [39–42]. A composition-only baseline (per-element statistics of atomic number, mass, electronegativity, atomic radius, plus group, period, transition-metal and rare-earth fractions) achieves 56.7% accuracy (60.7% balanced accuracy). Adding two tiers of structural features – broad, near-universal-coverage fields (space group number, crystal-system bucket, unit-cell volume, density, site count) and local-geometry features around magnetic sites (nearest magnetic–magnetic neighbour distance, magnetic coordination number, fraction of magnetic sites; computed from atomic positions and species only, never from the magnetic-moment vectors that generate the label itself) – raises this to 67.4% accuracy (68.0% balanced accuracy) on a stratified 28,769/7,193 train/test split (Supplementary Fig. 1). The local-geometry tier is computed for every source with a public structure endpoint (Materials Project 37,783/37,783, JARVIS 19,116/19,116, AFLOW 2,637/2,639, NOMAD 495/500, 2DMatPedia 1,862/1,862; OQMD 0/816 because its REST endpoint was unreachable). Computing it for MAGNDATA alone, whose CIFs carry atomic positions inline, would make "tier present" a proxy for the one source whose labels are experimental and 83% AFM; with full coverage the headline numbers are essentially unchanged (67.0% → 67.4% accuracy, 68.1% → 68.0% balanced), but per-source test

accuracy on Materials Project rises from 65.6% to 66.9% (balanced 63.0% → 65.9%), the inflated MAGNDATA figure drops from 89.5% to 75.8%, and four local-geometry features (mean/std/min magnetic–magnetic distance, mean magnetic coordination) enter the top seven features by LightGBM split importance, where before none appeared in the top ten. Space-group number remains the single most important feature, despite being absent from the baseline entirely. The classifier is retrained on all labeled data and deployed to predict ordering for every record lacking a ground-truth label, including all 19,116 JARVIS records, none of which carry a populated ordering field.

Graph neural network classifiers

Two graph-based models were trained on the train/test split of the LightGBM classifier and scored, together with it, on the 7,161 of its 7,193 test records whose structure is fully ordered (the other 32 are partially occupied MAGNDATA structures). Each crystal is a periodic graph whose nodes are atoms and whose edges join every atom to its 12 nearest neighbours within 8 Å, built from species, positions and lattice only; no magnetic moment enters, since the MAGNDATA labels are derived from the moments. The first is a crystal graph convolutional neural network (CGCNN) [43] trained from scratch in PyTorch Geometric [44]: a learned 64-dimensional element embedding projected to 128 channels, four gated graph convolutions with batch normalization and residual updates, bond lengths expanded in 40 Gaussians over 0–8 Å, mean pooling over atoms and a two-layer head. It is trained with class-weighted cross-entropy and AdamW (learning rate 10^{-3}, weight decay 10^{-4}, cosine schedule) for 120 epochs in batches of 128, keeping the epoch of best balanced accuracy on a stratified 10% validation fraction of the training set, with three seeds on one 24 GB GPU. The second probes a pretrained foundation potential: the invariant per-atom descriptors of both interaction layers of the frozen MACE-MP-0 `medium` model [45] (128 channels, `max_L`= 1, $r_{\max} = 6.0$ Å; distinct from the fine-tuned `small` model used for relaxations below) are pooled over atoms by mean and standard deviation into a 512-dimensional vector, which LightGBM, with the settings of the screening classifier, classifies alone and concatenated with the screening features. MACE-MP-0 is trained on energies, forces and stresses only, so the embedding carries no ordering label, but it is not independent of Materials Project: its MPtrj training set holds relaxation trajectories of the same structures,

computed in their stored magnetic orderings. Accuracy on the Materials Project rows measures agreement with Materials Project's label, which is the label this work finds biased towards FM; the MAGNDATA rows (311 AFM, 39 FiM, 19 FM in the test set) are the only experimental ground truth, and their balanced accuracy shifts by up to 4 points when the training set changes by only the 26 candidates and by 6 points between CGCNN seeds, as much as it differs between models. For an external test against DFT, every model was retrained with the 26 DFT-validated candidates removed from training, and its predictions for them compared with the DFT verdicts; with 21 overturned and 3 confirmed candidates these are reported as counts.

Screening funnel and shortlist

The noncentrosymmetric filter was verified computationally against all 230 space groups, recovering the textbook 92 centrosymmetric / 138 noncentrosymmetric split. A hull energy is known for 1,340 of the 1,864 heavy-SOC survivors (the MP, JARVIS and OQMD records), 671 of which lie within 100 meV atom^{-1} of it. The `ehull` field of the JARVIS-DFT snapshot used here is not a hull distance (Supplementary Note 2), so the hull distance of every JARVIS record was recomputed from JARVIS's own OptB88vdW formation energies with a per-chemical-system convex hull (13,967 of 19,116 records within 100 meV atom^{-1}); three of the four JARVIS entries in the original top-30 shortlist fail this criterion, and none entered the DFT batch, which is Materials Project only. The hull criterion is available both as a hard filter (records from sources that provide no hull energy – AFLOW, NOMAD, 2DMatPedia, MAGNDATA – are retained rather than penalised for a missing field; 1,195 candidates) and as a sort key; the DFT batch was drawn from the sort-key ranking, which is kept alongside so the batch remains reproducible.

For the top 100 screened candidates per source, structures were backfilled via each source's own public data-access method (298 of 300 attempted backfills succeeded; two AFLOW records failed on repeated TLS errors from the AFLUX endpoint) and re-relaxed with a MACE foundation-model potential [45, 46], fine-tuned on the harmonized dataset via multi-head training (a replay head sampling 20,000 configurations from the original pretraining distribution to avoid catastrophic forgetting), as an independent consistency check against each source's own DFT relaxation. The base model is MACE-MP-0 `small` (`num_channels`= 128,

max_L= 0, two interaction layers, $r_{\max} = 6.0$ Å); the relaxations use the Stage-2 stochastic-weight-averaging checkpoint of the fine-tune with its default head, and are static FIRE [47] optimizations in ASE [48]. Other universal potentials trained on Materials Project data, such as M3GNet [49] and CHGNet [50], would serve the same purpose. All 298 relaxations converged (max force ≤ 0.05 eV Å^{-1}) with a median relaxation-energy shift of 0.9 meV atom^{-1} relative to the source structure. Classifier confidence and MACE relaxation consistency were merged into a final ranking (converged → hull-stable → smallest relaxation shift → classifier confidence); the resulting top-30 shortlist is the pool the DFT batch is drawn from.

DFT ordering-validation protocol

For shortlisted candidates carrying an `ordering_source=ground_truth`, `confidence=1.0` label from Materials Project (i.e., MP's own recorded ordering, not a classifier prediction), we apply the collinear magnetic-ordering energy-mapping approach behind MP's own workflow [14]: symmetry-distinct FM and AFM spin configurations are enumerated with `pymatgen.MagneticStructureEnumerator` [51] (backed by `enumlib`); VASP inputs are generated with `MPRelaxSet` – MP's own PBE exchange-correlation functional [52], GGA+U parameters [53–55] (e.g. LDAUU= 5.3 eV on Fe), plane-wave cutoff (520 eV), and Monkhorst–Pack k-point convention [56] – overriding only the initial MAGMOM per configuration; and each configuration is relaxed to convergence within Kohn–Sham density-functional theory [57, 58] as implemented in VASP 6.4.3 [59] with the projector-augmented-wave method [60, 61] (`MPRelaxSet` defaults ISIF= 3, ISMEAR= −5, i.e. the tetrahedron method with Blöchl corrections [62], EDIFF= 5×10^{-5} eV per atom, i.e. 0.05 meV atom^{-1}). This keeps the energies directly comparable to MP's stored values rather than constituting an independent DFT recipe. Beyond VASP's own convergence flag, every calculation was audited in two further ways: the final RMS charge-density residual was checked (the "EDIFF is reached" banner can fire on a transient while the density is still far from self-consistent), and the final per-ion moments were compared against the intended MAGMOM pattern and against the moment magnitudes of the FM calculation of the same compound. A configuration whose magnetic ions collapse to a low-spin or nonmagnetic state, or whose moments re-order into the FM pattern during the self-consistent-field (SCF) cycle (a doubled "AFM" cell that is really FM), passes VASP's convergence test but is not a sample of the ordering

it was labeled with, and is excluded from the verdict rather than counted. As a check of the setup, the converged FM configuration of mp-861559 ($TaFeO_4$) reproduces MP's recorded total magnetization to within $0.003\,\mu_B$ (30.00000 vs. $30.0022\,\mu_B$ for 36 atoms), ruling out a convention mismatch, while three symmetry-distinct AFM configurations all converge below it, the lowest by 0.274 eV per cell (≈45.7 meV per Fe).

The first pass (93 relaxations across the 26 candidates; 116 audited relaxations in total once the second batch is included) left 12 calculations in seven candidates that failed the spin-state audit: collapsed or low-spin Fe/Cr moments in an AFM configuration (mp-1173139, mp-1173143, mp-551086, mp-775194), "AFM" cells whose moments re-ordered to the FM pattern during the SCF cycle (mp-682554), an "AFM" pattern that had flipped only a nonmagnetic d^0 Ti site and was therefore FM in disguise (mp-1216981), and an FM reference that collapsed to a nonmagnetic state (mp-22972). A second batch replaced every failed configuration with a freshly enumerated one in a spin state that the audit accepts: additional symmetry-distinct AFM/FiM cells for mp-1173139, mp-1173143, mp-775194 and mp-551086 (the ferrimagnetic ones with the two Fe sublattices, or Fe and Cr, unequal in number), Fe-only AFM supercells for mp-1216981 (Ti excluded from the enumeration), anti-collapse FM reruns for mp-22972 (in the primitive cell and in the cell of its lowest AFM configuration), and, for mp-682554, whose $1 \times 1 \times 2$ Fe–Fe stacking (130 atoms, 27.9 Å between the two Fe sites) diverged twice at the first density-mixing step through charge sloshing, five single-point mixer/smearing variants of the same cell run side by side, with the production relaxation restarted only under the first variant that converged; that production run was stopped before completion and is the one configuration missing from the batch. A superseded attempt (a diverged or collapsed run that was re-done) is neither a sample nor a gap in the count.

Two protocol effects matter at the sub-meV atom^{-1} level. First, the MP recipe compares an FM relaxation in the primitive cell with AFM relaxations in supercells, each with its own k-mesh and each relaxed with ISIF= 3 to its own cell shape; for the sixteen configurations run both ways this "loose" margin differs from a "tight" one (both orderings relaxed with ISIF= 2 in the same relaxed AFM cell, identical k-mesh, EDIFF= 10^{-6} eV, EDIFFG= -0.01 eV Å^{-1}) by up to 1.3 meV atom^{-1} in either direction ($Ba_4Ta_{10}FeO_{30}$: $+1.18 \rightarrow 0.00$; $FeP_6(WO_8)_3$: $+2.09 \rightarrow +3.38$). Every candidate whose verdict rested on a loose margin smaller than about 2 meV atom^{-1} was therefore re-decided with a tight pair built in the cell of its lowest valid non-FM configuration (14 pairs in the first tight batch, 6 more for the remaining

small-margin overturns), and a tight number replaces the loose one for that configuration wherever it exists. Second, the verdict rules are asymmetric by construction: a label is *overturned* as soon as one audited configuration lies more than 0.10 meV atom^{-1} below FM – a still-missing configuration cannot undo that – whereas *confirmed FM* and *degenerate* require every staged configuration to have a valid energy. The 0.10 meV atom^{-1} window is twice the per-atom EDIFF of the loose protocol. The two 130-atom tight relaxations of mp-682554, first cut off short of the force criterion, were continued to convergence (29 and 32 ionic steps, final maximum forces 0.006 and 0.010 eV Å^{-1}); the gap moved from 0.315 to 0.313 meV atom^{-1} (40.7 meV per cell).

All five tungsten-bearing candidates initially failed to converge under MP's standard Hubbard correction (LDAUU= 6.2 eV on W): tungsten in these oxides is typically W^{6+}, a nominally d^0 ion, and a large U applied to a near-empty d shell leaves the self-consistent density with no restoring force, oscillating indefinitely rather than converging. Removing U from tungsten (LDAUL= -1), applied uniformly across every ordering of the affected compound, converged all 18 corresponding production calculations on the first retry. One of the five is not d^0: in $FeP_6(WO_8)_3$ (mp-850225), charge balance with Fe^{3+} and P^{5+} requires W^{5+} (d^1), and in the U-free runs each W carries 0.78–0.86 μ_B antiparallel to Fe in the FM state. These reruns also fix the cell shape and volume at MP's relaxed values (ISIF= 2, ions only) and use Gaussian smearing (ISMEAR= 0, $\sigma = 0.05$ eV; tetrahedron smearing for mp-755203) for SCF stability. This changes the absolute energy scale for those five candidates relative to MP's U-inclusive value, but not the FM-versus-AFM comparison within each compound, since the same settings are applied to every ordering of that compound.

Exchange and DMI of mp-754090

Effective spin-Hamiltonian parameters are extracted by energy mapping, the standard route from the total energies of constructed spin configurations to Heisenberg parameters [63–65]. The isotropic exchange is obtained from the tight same-cell FM/AFM pair used for the ordering verdict (both orderings relaxed in the same cell, which is the fixed lattice a Heisenberg J is defined on). Each of the two Fe sites (Fe0, Fe1) per primitive cell of the rock-salt-derived $Li_4Fe_2TeWO_{12}$ structure has twelve Fe neighbours, six in a triangular layer (5.112–5.157 Å, bond angles of 60°, 120° and 180°; four on the opposite sublattice and

two on the same sublattice by lattice translation) and six across the 4.53 Å layer spacing (5.411–5.427 Å; again four on the opposite sublattice and two on the same). The next shell starts at 7.45 Å. For $H = -\sum_{\langle ij \rangle} J\, \hat{s}_i \cdot \hat{s}_j$ with each bond counted once, the collinear AFM configuration (Fe0 sublattice reversed) flips the eight inter-sublattice bonds per cell – four in-plane and four inter-layer – so $E_{\mathrm{AFM}} - E_{\mathrm{FM}} = 2 \times 8\, J = 16J$; with the same-cell DFT value $\Delta E = 44.54$ meV per two-Fe cell this gives $J = \Delta E/16 = 2.78$ meV per bond. The loose pair (each ordering at its own ISIF= 3 optimum) gives $\Delta E = 48.23$ meV and $J = 3.01$ meV, an 8% protocol spread that we carry as the systematic uncertainty of J. A single FM/AFM pair cannot separate the in-plane from the inter-layer exchange, nor resolve the four same-sublattice bonds, which are invisible to a two-configuration fit; all twelve bonds are assigned the same J in the spin model – an explicit assumption. This is a single-J effective model, not a multi-shell decomposition; its $zJ = 12J = 33.4$ meV, which fixes the mean-field ordering temperature, is what the DFT pair determines directly.

DMI does not appear in a collinear FM-versus-AFM comparison; it requires spins canted relative to one another. The Li/Fe/Te/W cation ordering reduces the space group to $P1$, so no inversion centre at a bond midpoint forbids it [2]. We ran two static, constrained noncollinear-spin+SOC calculations from the relaxed FM structure – the established route to a first-principles DMI [66, 67] – within unconstrained noncollinear PAW magnetism [68], using the penalty-energy method [69] (`I_CONSTRAINED_M=1`, constraining moment direction only) to hold Fe0 along $+\hat{z}$ and Fe1 along $+\hat{x}$ (canted$_+$) or $-\hat{x}$ (canted$_-$) – i.e. $\theta = \pm 90°$ canting in the xz-plane, an arbitrary but fixed reference axis since $P1$ symmetry privileges no plane. The constraint held essentially exactly (Fe1 converged to $4.038\,\mu_B$ along x in canted$_+$ and $-4.039\,\mu_B$ in canted$_-$; Fe0 leakage off z was $< 0.002\,\mu_B$ in both). With the same bond-counting convention as for J ($z = 8$ inter-sublattice bonds per cell, each counted once; the four same-sublattice bonds connect parallel spins and contribute to neither term), $E(\theta) = E_0 - zJ\cos\theta - zD\sin\theta$, and comparing $\theta = \pm 90°$ isolates the antisymmetric term since $\cos(\pm 90°) = 0$:

$$zD = \frac{E(\theta = -90°) - E(\theta = +90°)}{2}, \tag{1}$$

giving $E(-90°) - E(+90°) = 1.546$ meV per cell and $D = 0.097$ meV per bond, i.e. $D/J \approx 3.5\%$. Because $\hat{s}_1 = \hat{z}$ and the canting is confined to the xz-plane, $\hat{s}_1 \times \hat{s}_2 = (0, \sin\theta, 0)$: this probe measures the mean y-component of the DMI vectors of the eight inequivalent inter-sublattice bonds in this lab frame, not a generic magnitude, and is a resolved DFT

number for one projection of D rather than a fully characterized three-dimensional DMI vector; a complete characterization would require at least eight independent canting probes.

Exchange and DMI of mp-1079285

The Fe atoms of Tb_6FeBi_2 form chains along b (Fe–Fe 4.198 Å), the next Fe–Fe distances being 8.396 Å along $\pm 2b$ and 8.396–8.407 Å to the six chain neighbours along $\pm a$, $\pm c$, $\pm(a+c)$. Tb is treated with the $4f$ electrons frozen in the core (`Tb_3` PAW), so the only moment in the model is Fe ($1.68\,\mu_B$ in the FM calculation), an explicit approximation that ignores any Tb–Fe exchange. Every collinear AFM configuration that the enumeration produced for this candidate turned out to be the same chain-antiparallel order in a different cell setting, so the tight pair in the $1 \times 2 \times 1$ cell resolves only the intra-chain coupling: with $H = -\sum_{\langle ij \rangle} J_{ij}\, \hat{s}_i \cdot \hat{s}_j$ (each bond counted once) it flips two chain bonds per cell, $E_{\mathrm{AFM}} - E_{\mathrm{FM}} = 4J_b$. A chain model alone has no order at any $T > 0$, so three further same-cell tight pairs were staged in supercells of the relaxed FM cell (`ISIF`= 2, `EDIFF`= 10^{-6} eV, `EDIFFG`= -0.01 eV Å^{-1}): the $2a$ and $2c$ cells with the two Fe antiparallel, which flip the $\pm a, \pm(a+c)$ and the $\pm c, \pm(a+c)$ bonds respectively ($\Delta E_{2a} = 4(J_a + J_{ac})$, $\Delta E_{2c} = 4(J_c + J_{ac})$), and a $1\times4\times1$ cell with the chain pattern $++--$, which flips two of the four chain bonds and all four $\pm 2b$ bonds ($\Delta E_{4b} = 4J_b + 8J_{2b}$). Two equations do not determine three inter-chain couplings; the model takes their mean, $J_\perp = (\Delta E_{2a} + \Delta E_{2c})/16$, under the stated assumption $J_a \approx J_c \approx J_{ac}$ (the three bond lengths agree to 0.01 Å). The $\pm 2b$ bond is resolved separately because its length coincides with the inter-chain shell, which no radial exchange function in the spin-dynamics code can separate. The J_b and ΔE_{4b} pairs use different k-meshes ($3\times3\times3$ on the $1 \times 2 \times 1$ cell, $3 \times 2 \times 3$ on the $1 \times 4 \times 1$ cell) and cells: the $1 \times 2 \times 1$ pair sits in the relaxed cell of the enumerated AFM configuration ($a = 8.372$, $c = 8.376$ Å) with a mesh equivalent to 6 points along b in the primitive cell, the $1 \times 4 \times 1$ pairs in the relaxed FM cell ($a = 8.407$, $c = 8.396$ Å) with 8. The same-cell $+-+-$ check is compared with the FM member of the $++--$ pair; its relaxation, first cut off at ionic step 24, was continued to the force criterion (28 ionic steps, final maximum force 0.007 eV Å^{-1}), lowering ΔE by 0.48 meV from the cut-off value.

The Pm mirror plane is normal to b and contains the Fe site, so by Moriya's rules $\mathbf{D} \perp b$ and two projections determine $|\mathbf{D}|$. Each was obtained from a pair of 90° spin spirals in the

$1 \times 4 \times 1$ cell (four Fe, moments constrained with `I_CONSTRAINED_M`= 1, $\lambda = 10$, SOC on, `EDIFF`= 10^{-7} eV, single points on the relaxed FM structure), rotating in the plane normal to $\hat{n}_1 \parallel a$ or $\hat{n}_2 = \hat{b} \times \hat{n}_1$ with either chirality. For a 90° spiral $\hat{s}_i \times \hat{s}_j = \pm\hat{n}$ on every chain bond while the symmetric exchange (including J_{2b}, whose bonds are antiparallel in both chiralities) is identical for the two, so $\mathbf{D}\cdot\hat{n} = [E(+) - E(-)]/8$ per bond. The constraint penalty energies were $\leq 1.6 \times 10^{-9}$ eV, the Fe moments had no component off their target axes (to the printed $0.001\,\mu_B$) with $|m_{\mathrm{Fe}}| = 1.50 \pm 0.01\,\mu_B$ in all four calculations, and the induced Tb moments ($\leq 0.123\,\mu_B$; Bi $\leq 0.002\,\mu_B$) rotate with the spiral, their magnitudes agreeing between chiralities to $0.001\,\mu_B$.

Exchange of mp-850225 and the unresolved spiral

$FeP_6(WO_8)_3$ has one Fe per rhombohedral cell, so its Fe sublattice is a Bravais lattice on which a two-sublattice canting probe sees no DMI (the $+\mathbf{t}$ and $-\mathbf{t}$ bonds contribute opposite $\mathbf{D} \cdot (\mathbf{S}_0 \times \mathbf{S}_1)$). The two exchange shells are separated by two same-cell pairs in 68-atom supercells: the first, from the ordering enumeration, gives $\Delta E = 8(J_1 + J_2)$; the second, with alternate hexagonal Fe layers antiparallel along c, every out-of-plane bond and no in-plane one ($\Delta E = 12J_2$). Resolving D needs a spin spiral in the 102-atom hexagonal cell. That spiral pair (noncollinear with SOC, Fe directions constrained, W moments free) was run in three attempts (about 65 node-hours): one chirality converged (EDIFF = 10^{-6} eV), but the other never reached self-consistency at the precision the extraction needs ($\Delta E = 18 \sin 120° D_z \approx 15.6\, D_z$, i.e. about 1.6 meV for $D \sim 0.1$ meV), so no DMI is quoted for this compound.

Finite-temperature spin dynamics

The DFT-derived $J = 2.78$ meV, with no fitting or rescaling, parameterizes a twelve-neighbour Heisenberg Hamiltonian for the Fe sublattice of mp-754090 (framework Li/Te/W/O sites carry no spin and are omitted from the geometry), implemented in LAMMPS-SPIN [32] within LAMMPS [70, 71] via the `spin/exchange` pair style (which counts each bond once, matching the convention of the DFT extraction) with a 5.5 Å cutoff that captures all twelve Fe–Fe bonds per site. `spin/exchange` uses the Bethe–Slater

radial form $J(r) = 4a\,(r/d)^2 e^{-(r/d)^2}$ (with its b parameter set to zero); $d = 5.274$ Å is solved so that $J(5.130\ \text{Å}) = J(5.418\ \text{Å})$ exactly and a set so that both equal the DFT J, which makes $J(r)$ constant to within 0.06% over the two shells. A zero-step energy evaluation of the FM and Fe0-reversed configurations of the model reproduces the DFT $E_{\text{AFM}} - E_{\text{FM}} = 16J = 44.54$ meV per cell to 0.003%. The DMI is not included in the production sweep: the LAMMPS `spin/dmi` pair style implements $\mathbf{D}_{ij} = D\,(\hat{e}_{ij} \times \hat{v})$ with a single global vector $\hat{v}$, and for the DFT probe geometry ($\hat{s}_i \times \hat{s}_j = \pm\hat{y}$) its energy difference is $D\,(\hat{v} \times \hat{y}) \cdot \sum_b \hat{e}_b$ over the eight inter-sublattice bond directions – a sum that is 0.03 (in units of a bond vector) for this near-centrosymmetric neighbour shell, so no choice of $\hat{v}$ can reproduce the DFT $E(-90°) - E(+90°)$ (a zero-step evaluation with $D = 0.097$ meV and $\hat{v} = \hat{y}$ gives 0.000 meV); a faithful model needs the eight bond-resolved DMI vectors. We instead bound the effect of a DMI of the measured magnitude on $|m|(T)$ with a control sweep that adds `spin/dmi` with $D = 0.097$ meV and $\hat{v} = \hat{y}$ (an arbitrary but representative placement of a 3.5% antisymmetric term) at four temperatures. The model is run on a $16 \times 8 \times 8$ replication (2048 spins) of the two-atom Fe primitive cell, with periodic boundaries, a rigid (spin-only) lattice and zero single-ion anisotropy. Each run starts from the fully aligned FM ground state and is evolved under Langevin spin dynamics ($dt = 0.1$ fs) – stochastic Landau–Lifshitz–Gilbert dynamics of classical spins [72, 73], the standard atomistic approach to finite-temperature magnetism [74, 75] – for 200,000 steps (20 ps) at twelve temperatures from 10 to 400 K; the normalized net magnetization $|m| = |\sum_i \hat{s}_i|/N$ is averaged over the second half of each run and over three independent thermostat seeds per temperature. The Langevin damping is $\alpha = 1$: the equilibrium $|m|(T)$ does not depend on α, but the thermalization time of the softest magnons of the finite cell scales as $1/\alpha$ (see Spin-model validation checks). The independent Metropolis Monte Carlo runs 2000 equilibration and 2000 measurement sweeps per temperature on the same 2048-spin lattice.

For mp-1079285 the same machinery carries a three-shell model,

$$H = -\Big(J_b \sum_{\text{chain}} + J_{2b} \sum_{\pm 2b} + J_\perp \sum_{\text{inter}}\Big)\hat{s}_i \cdot \hat{s}_j + \sum_{\text{chain}} \mathbf{D} \cdot (\hat{s}_i \times \hat{s}_j), \tag{2}$$

on the Fe sublattice (one spin per primitive cell, $|m| = 1.68\,\mu_B$ as a classical unit vector). Two `spin/exchange` instances carry the chain bond (cutoff 5 Å) and the 8.4 Å shell (Bethe–Slater radial form with its root placed at the chain distance); because the $\pm 2b$ bond has the same length as the inter-chain bonds, spins carry an atom type equal to their chain index

modulo four, so that $\pm 2b$ pairs (types differing by two) receive J_{2b} by an explicit per-type override while inter-chain pairs (equal types) receive $J_\perp$. The DMI is carried by `spin/dmi` on the chain bond only (5 Å cutoff), with its vector calibrated – not assumed – by zero-step energy evaluations of the very 90° spirals used in the DFT, so that the model reproduces $E(+) - E(-)$ for both spiral planes. Before any sweep, zero-step evaluations of the $1 \times 2 \times 1$, $1 \times 4 \times 1$ $(++--)$, $2a$ and $2c$ cells reproduce the four DFT energy differences – the two chain cells exactly, the $2a$ and $2c$ cells to 0.5%, which is the residual of fitting one $J_\perp$ to two inequivalent shells. The production cell is an $8 \times 8 \times 8$ replication of the $1 \times 4 \times 1$ unit (2048 spins), started from FM along b, with the same Langevin protocol at fourteen temperatures from 5 to 300 K. The same-cell parametrization was swept at 10, 100, 125, 150 and 200 K with three seeds each.

Finite-size scaling

Sweeps were repeated at $N = 256$, 864, 2048 and 4000 spins with the box aspect ratio held fixed, so that every size sees the same neighbour shells, as 40 ps Langevin runs ($\alpha = 1$) with the moments pooled over the last 20 ps of four thermostat seeds; for mp-1079285 the 135–150 K window, where the crossings lie, was run with 10–14 seeds per point, six of them 80 ps. Error bars on U_4 are a drop-one-seed jackknife. U_4 of a three-component order parameter tends to 2/3 in the ordered phase and 4/9 in the disordered one and is size-independent at T_c. Because U_4 is a ratio of small fluctuating moments, its jackknife error grows from ± 0.0003–0.002 at 95 K to ± 0.03 above T_c for mp-754090 (from ± 0.0002–0.0007 at 120 K to ± 0.004–0.028 at 150 K for mp-1079285), and in that noisy regime the difference between two sizes can change sign on scales far below its own error bar. The quoted crossings are therefore the lowest-temperature sign changes that persist: the difference is required to stay positive at a majority of the higher temperatures measured, which discards both the sub-noise flips inside the ordered plateau, where every U_4 has converged on 2/3, and isolated single-point reversals above T_c; one such non-persistent sign change was discarded for mp-1079285, whose 155 and 160 K points retain only four seeds each and scatter back above 4/9. T_c and its uncertainty are the mean and standard deviation of the three pairwise crossings of consecutive sizes. The longer runs at 135–150 K cut the U_4 error where the crossings lie ($\pm 0.048 \to \pm 0.014$ for $N = 4000$ at 145 K), though not uniformly: the same cell's error at

150 K grew from ±0.020 to ±0.028 as the longer runs sampled more of the disordered-phase excursions.

Re-deriving both crossings under changes to the sampling window (Supplementary Note 3) shows that the half-run equilibration cut is required and that the autocorrelation time of U_4 near T_c is comparable to the run length, so the quoted spreads are lower bounds.

Spin-model validation checks

Each model passes four checks before any sweep (Supplementary Note 4 describes the errors they caught): a zero-step energy evaluation of the model against every DFT pair it was built from (for mp-1079285 the $++--$ check is what exposes a $\pm 2b$ bond silently given $J_\perp$); an explicit count and geometry of the neighbour shells the cutoff admits; an independent Metropolis Monte Carlo calculation of $|m|(T)$ on the same lattice; and a window-drift test on every reported point, comparing the mean of $|m|$ over consecutive 2 ps windows of the averaged half. For the mp-1079285 production sweep the seed-averaged drift is ≤ 0.008 at every temperature except +0.018 at 150 K, against within-run fluctuations of ±0.05 there, with no monotonic trend; for the same-cell sweep it is ≤ 0.008 except +0.05 at 150 K, the transition point, against a within-run fluctuation of ±0.04; for mp-754090 the 100 K point has a within-run fluctuation of 0.043 and window means that differ by up to 0.049.

DATA AVAILABILITY

The per-candidate DFT verdict tables, the spin-model parameter files and the reduced magnetization and finite-size-scaling sweeps behind every figure are available at https://github.com/akhtar07/magnetic-materials-dmi-screening and are provided as Source Data with this paper. The harmonized database is rebuilt from the seven public sources by the code below.

CODE AVAILABILITY

The database harmonization, classifier (LightGBM and graph-neural-network), screening, VASP-input generation, spin-model, spin-dynamics, finite-size-scaling and figure scripts are

available at https://github.com/akhtar07/magnetic-materials-dmi-screening.

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AUTHOR CONTRIBUTIONS

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COMPETING INTERESTS

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[1] I. Dzyaloshinsky, A thermodynamic theory of "weak" ferromagnetism of antiferromagnetics, J. Phys. Chem. Solids **4**, 241 (1958).

[2] T. Moriya, Anisotropic superexchange interaction and weak ferromagnetism, Phys. Rev. **120**, 91 (1960).

[3] A. Bogdanov and A. Hubert, Thermodynamically stable magnetic vortex states in magnetic crystals, J. Magn. Magn. Mater. **138**, 255 (1994).

[4] A. Fert, V. Cros, and J. Sampaio, Skyrmions on the track, Nat. Nanotechnol. **8**, 152 (2013).

[5] N. Nagaosa and Y. Tokura, Topological properties and dynamics of magnetic skyrmions, Nat. Nanotechnol. **8**, 899 (2013).

[6] S. Mühlbauer, B. Binz, F. Jonietz, C. Pfleiderer, A. Rosch, A. Neubauer, R. Georgii, and P. Böni, Skyrmion Lattice in a Chiral Magnet, Science **323**, 915 (2009).

[7] X. Z. Yu, Y. Onose, N. Kanazawa, J. H. Park, J. H. Han, Y. Matsui, N. Nagaosa, and Y. Tokura, Real-space observation of a two-dimensional skyrmion crystal, Nature **465**, 901 (2010).

[8] S. Heinze, K. von Bergmann, M. Menzel, J. Brede, A. Kubetzka, R. Wiesendanger, G. Bihlmayer, and S. Blügel, Spontaneous atomic-scale magnetic skyrmion lattice in two dimensions, Nat. Phys. **7**, 713 (2011).

[9] N. Romming, C. Hanneken, M. Menzel, J. E. Bickel, B. Wolter, K. von Bergmann, A. Kubetzka, and R. Wiesendanger, Writing and Deleting Single Magnetic Skyrmions, Science **341**, 636 (2013).

[10] S. Woo, K. Litzius, B. Krüger, *et al.*, Observation of room-temperature magnetic skyrmions and their current-driven dynamics in ultrathin metallic ferromagnets, Nat. Mater. **15**, 501 (2016).

[11] A. Fert, N. Reyren, and V. Cros, Magnetic skyrmions: advances in physics and potential applications, Nat. Rev. Mater. **2**, 17031 (2017).

[12] Y. Tokura and N. Kanazawa, Magnetic skyrmion materials, Chem. Rev. **121**, 2857 (2021).

[13] A. Jain, S. P. Ong, G. Hautier, W. Chen, W. D. Richards, S. Dacek, S. Cholia, D. Gunter, D. Skinner, G. Ceder, and K. A. Persson, Commentary: The Materials Project: A materials genome approach to accelerating materials innovation, APL Mater. **1**, 011002 (2013).

[14] M. K. Horton, J. H. Montoya, M. Liu, and K. A. Persson, High-throughput prediction of the ground-state collinear magnetic order of inorganic materials using Density Functional Theory, npj Comput. Mater. **5**, 64 (2019).

[15] A. E. Fahmy, Learning Magnetic Order Classification from Large-Scale Materials Databases, arXiv:2509.05909v2 (2026).

[16] S. V. Gallego, J. M. Perez-Mato, L. Elcoro, E. S. Tasci, R. M. Hanson, K. Momma, and M. I. Aroyo, MAGNDATA: towards a database of magnetic structures. I. The commensurate case, J. Appl. Crystallogr. **49**, 1750 (2016).

[17] S. V. Gallego, J. M. Perez-Mato, L. Elcoro, E. S. Tasci, R. M. Hanson, and G. Madariaga, MAGNDATA: towards a database of magnetic structures. II. The incommensurate case, J. Appl. Crystallogr. **49**, 1941 (2016).

[18] K. Choudhary, K. F. Garrity, A. C. E. Reid, *et al.*, The joint automated repository for various integrated simulations (JARVIS) for data-driven materials design, npj Comput. Mater. **6**, 173

(2020).

[19] S. Curtarolo, W. Setyawan, G. L. W. Hart, *et al.*, AFLOW: An automatic framework for high-throughput materials discovery, Comput. Mater. Sci. **58**, 218 (2012).

[20] J. Zhou, L. Shen, M. D. Costa, *et al.*, 2DMatPedia, an open computational database of two-dimensional materials from top-down and bottom-up approaches, Sci. Data **6**, 86 (2019).

[21] J. E. Saal, S. Kirklin, M. Aykol, B. Meredig, and C. Wolverton, Materials Design and Discovery with High-Throughput Density Functional Theory: The Open Quantum Materials Database (OQMD), JOM **65**, 1501 (2013).

[22] S. Kirklin, J. E. Saal, B. Meredig, A. Thompson, J. W. Doak, M. Aykol, S. Rühl, and C. Wolverton, The Open Quantum Materials Database (OQMD): assessing the accuracy of DFT formation energies, npj Comput. Mater. **1**, 15010 (2015).

[23] C. Draxl and M. Scheffler, NOMAD: The FAIR concept for big data-driven materials science, MRS Bull. **43**, 676 (2018).

[24] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, LightGBM: A highly efficient gradient boosting decision tree, Adv. Neural Inf. Process. Syst. **30** (2017).

[25] A. Fert and P. M. Levy, Role of anisotropic exchange interactions in determining the properties of spin-glasses, Phys. Rev. Lett. **44**, 1538 (1980).

[26] A. Crépieux and C. Lacroix, Dzyaloshinsky–Moriya interactions induced by symmetry breaking at a surface, J. Magn. Magn. Mater. **182**, 341 (1998).

[27] H. Yang, A. Thiaville, S. Rohart, A. Fert, and M. Chshiev, Anatomy of Dzyaloshinskii-Moriya interaction at Co/Pt interfaces, Phys. Rev. Lett. **115**, 267210 (2015).

[28] P. W. Anderson, Antiferromagnetism. Theory of Superexchange Interaction, Phys. Rev. **79**, 350 (1950).

[29] J. B. Goodenough, Theory of the role of covalence in the perovskite-type manganites [La,M(II)]MnO_3, Phys. Rev. **100**, 564 (1955).

[30] J. Kanamori, Superexchange interaction and symmetry properties of electron orbitals, J. Phys. Chem. Solids **10**, 87 (1959).

[31] M. Hellenbrandt, The Inorganic Crystal Structure Database (ICSD)—Present and Future, Crystallogr. Rev. **10**, 17 (2004).

[32] J. Tranchida, S. J. Plimpton, P. Thibaudeau, and A. P. Thompson, Massively parallel symplectic algorithm for coupled magnetic spin dynamics and molecular dynamics, J. Comput.

Phys. **372**, 406 (2018).

[33] N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, and E. Teller, Equation of State Calculations by Fast Computing Machines, J. Chem. Phys. **21**, 1087 (1953).

[34] K. Binder, Finite size scaling analysis of ising model block distribution functions, Z. Phys. B **43**, 119 (1981).

[35] K. Chen, A. M. Ferrenberg, and D. P. Landau, Static critical behavior of three-dimensional classical Heisenberg models: A high-resolution Monte Carlo study, Phys. Rev. B **48**, 3249 (1993).

[36] S. P. Ong, S. Cholia, A. Jain, M. Brafman, D. Gunter, G. Ceder, and K. A. Persson, The Materials Application Programming Interface (API): A simple, flexible and efficient API for materials data based on REpresentational State Transfer (REST) principles, Comput. Mater. Sci. **97**, 209 (2015).

[37] L. Ward, A. Agrawal, A. Choudhary, and C. Wolverton, A general-purpose machine learning framework for predicting properties of inorganic materials, npj Comput. Mater. **2**, 16028 (2016).

[38] J. Schmidt, M. R. G. Marques, S. Botti, and M. A. L. Marques, Recent advances and applications of machine learning in solid-state materials science, npj Comput. Mater. **5**, 83 (2019).

[39] J. Nelson and S. Sanvito, Predicting the Curie temperature of ferromagnets using machine learning, Phys. Rev. Mater. **3**, 104405 (2019).

[40] C. J. Court and J. M. Cole, Magnetic and superconducting phase diagrams and transition temperatures predicted using text mining and machine learning, npj Comput. Mater. **6**, 18 (2020).

[41] N. C. Frey, M. K. Horton, J. M. Munro, S. M. Griffin, K. A. Persson, and V. B. Shenoy, High-throughput search for magnetic and topological order in transition metal oxides, Sci. Adv. **6**, eabd1076 (2020).

[42] A. Kabiraj, M. Kumar, and S. Mahapatra, High-throughput discovery of high Curie point two-dimensional ferromagnetic materials, npj Comput. Mater. **6**, 35 (2020).

[43] T. Xie and J. C. Grossman, Crystal graph convolutional neural networks for an accurate and interpretable prediction of material properties, Phys. Rev. Lett. **120**, 145301 (2018).

[44] M. Fey and J. E. Lenssen, Fast graph representation learning with PyTorch Geometric, arXiv:1903.02428 (2019).

[45] I. Batatia, P. Benner, Y. Chiang, *et al.*, A foundation model for atomistic materials chemistry, arXiv:2401.00096 (2023).

[46] I. Batatia, D. P. Kovács, G. N. C. Simm, C. Ortner, and G. Csányi, MACE: Higher order equivariant message passing neural networks for fast and accurate force fields, Adv. Neural Inf. Process. Syst. **35** (2022).

[47] E. Bitzek, P. Koskinen, F. Gähler, M. Moseler, and P. Gumbsch, Structural Relaxation Made Simple, Phys. Rev. Lett. **97**, 170201 (2006).

[48] A. Hjorth Larsen, J. Jørgen Mortensen, J. Blomqvist, *et al.*, The atomic simulation environment—a Python library for working with atoms, J. Phys.: Condens. Matter **29**, 273002 (2017).

[49] C. Chen and S. P. Ong, A universal graph deep learning interatomic potential for the periodic table, Nat. Comput. Sci. **2**, 718 (2022).

[50] B. Deng, P. Zhong, K. Jun, J. Riebesell, K. Han, C. J. Bartel, and G. Ceder, CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling, Nat. Mach. Intell. **5**, 1031 (2023).

[51] S. P. Ong, W. D. Richards, A. Jain, *et al.*, Python Materials Genomics (pymatgen): A robust, open-source python library for materials analysis, Comput. Mater. Sci. **68**, 314 (2013).

[52] J. P. Perdew, K. Burke, and M. Ernzerhof, Generalized Gradient Approximation Made Simple, Phys. Rev. Lett. **77**, 3865 (1996).

[53] S. L. Dudarev, G. A. Botton, S. Y. Savrasov, C. J. Humphreys, and A. P. Sutton, Electron-energy-loss spectra and the structural stability of nickel oxide: An LSDA+U study, Phys. Rev. B **57**, 1505 (1998).

[54] L. Wang, T. Maxisch, and G. Ceder, Oxidation energies of transition metal oxides within the GGA+U framework, Phys. Rev. B **73**, 195107 (2006).

[55] A. Jain, G. Hautier, S. P. Ong, C. J. Moore, C. C. Fischer, K. A. Persson, and G. Ceder, Formation enthalpies by mixing GGA and GGA+U calculations, Phys. Rev. B **84**, 045115 (2011).

[56] H. J. Monkhorst and J. D. Pack, Special points for Brillouin-zone integrations, Phys. Rev. B **13**, 5188 (1976).

[57] P. Hohenberg and W. Kohn, Inhomogeneous Electron Gas, Phys. Rev. **136**, B864 (1964).

[58] W. Kohn and L. J. Sham, Self-Consistent Equations Including Exchange and Correlation Effects, Phys. Rev. **140**, A1133 (1965).

[59] G. Kresse and J. Furthmüller, Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set, Phys. Rev. B **54**, 11169 (1996).

[60] P. E. Blöchl, Projector augmented-wave method, Phys. Rev. B **50**, 17953 (1994).

[61] G. Kresse and D. Joubert, From ultrasoft pseudopotentials to the projector augmented-wave method, Phys. Rev. B **59**, 1758 (1999).

[62] P. E. Blöchl, O. Jepsen, and O. K. Andersen, Improved tetrahedron method for Brillouin-zone integrations, Phys. Rev. B **49**, 16223 (1994).

[63] A. I. Liechtenstein, M. I. Katsnelson, V. P. Antropov, and V. A. Gubanov, Local spin density functional approach to the theory of exchange interactions in ferromagnetic metals and alloys, J. Magn. Magn. Mater. **67**, 65 (1987).

[64] H. J. Xiang, E. J. Kan, S.-H. Wei, M. Whangbo, and X. G. Gong, Predicting the spin-lattice order of frustrated systems from first principles, Phys. Rev. B **84**, 224429 (2011).

[65] H. Xiang, C. Lee, H.-J. Koo, X. Gong, and M.-H. Whangbo, Magnetic properties and energy-mapping analysis, Dalton Trans. **42**, 823 (2013).

[66] M. Heide, G. Bihlmayer, and S. Blügel, Dzyaloshinskii-Moriya interaction accounting for the orientation of magnetic domains in ultrathin films: Fe/W(110), Phys. Rev. B **78**, 140403 (2008).

[67] F. Freimuth, S. Blügel, and Y. Mokrousov, Berry phase theory of Dzyaloshinskii–Moriya interaction and spin–orbit torques, J. Phys.: Condens. Matter **26**, 104202 (2014).

[68] D. Hobbs, G. Kresse, and J. Hafner, Fully unconstrained noncollinear magnetism within the projector augmented-wave method, Phys. Rev. B **62**, 11556 (2000).

[69] P.-W. Ma and S. L. Dudarev, Constrained density functional for noncollinear magnetism, Phys. Rev. B **91**, 054420 (2015).

[70] S. Plimpton, Fast Parallel Algorithms for Short-Range Molecular Dynamics, J. Comput. Phys. **117**, 1 (1995).

[71] A. P. Thompson, H. M. Aktulga, R. Berger, *et al.*, LAMMPS - a flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales, Comput. Phys. Commun. **271**, 108171 (2022).

[72] T. L. Gilbert, A phenomenological theory of damping in ferromagnetic materials, IEEE Trans. Magn. **40**, 3443 (2004).

[73] P.-W. Ma and S. L. Dudarev, Langevin spin dynamics, Phys. Rev. B **83**, 134418 (2011).

[74] B. Skubic, J. Hellsvik, L. Nordström, and O. Eriksson, A method for atomistic spin dynamics simulations: implementation and examples, J. Phys.: Condens. Matter **20**, 315203 (2008).

[75] R. F. L. Evans, W. J. Fan, P. Chureemart, T. A. Ostler, M. O. A. Ellis, and R. W. Chantrell, Atomistic spin model simulations of magnetic nanomaterials, J. Phys.: Condens. Matter **26**, 103202 (2014).

SUPPLEMENTARY INFORMATION

Supplementary Note 1: MAGNDATA retrieval

A new end-to-end scraper against the live Bilbao Crystallographic Server index recovered 1,919 of 2,414 entries (79.5%). The 495 failures break down as 449 HTTP 404s from the server's temporary `.mcif` endpoint, 41 structures the parser refuses (36 explicitly disordered, 5 incommensurate), 3 read timeouts, and 2 entries whose fetched CIF contained no structure. Within the 404s, 148 are entries whose formula carries fractional site occupancies and 3 carry HTML markup in the upstream formula string, which makes the request URL malformed; those 151, together with the 41 refusals, are the at most 192 entries attributable to a data-format limitation. The remaining 298 are 404s on well-formed URLs for integer-stoichiometry entries, and they are transient rather than permanent: they persist across the scraper's four in-run retries, but re-requesting one of them (entry 0.57, $ScFeO_3$) returned 404 and then 200 within minutes, while its entry page is served throughout. 79.5% is therefore a property of this run rather than a ceiling. Nothing below depends on the missing entries, all of which are simply absent from the training and validation sets, but the MAGNDATA ground truth used here is smaller than the source allows.

Supplementary Note 2: the JARVIS-DFT `ehull` field

The `ehull` field of the JARVIS-DFT snapshot used here is not a hull distance (its median over the 93,902-entry snapshot is 1.7 eV atom^{-1}, the known stable compound TbMnSi carries `ehull`= 2.45 eV atom^{-1}, and 101 of the 3,206 harmonized JARVIS records with a *positive* formation energy carry `ehull`= 0, among them $BaFeHg_4$ at +0.36 eV atom^{-1}); it was therefore discarded and the hull distance of every JARVIS record recomputed from JARVIS's own OptB88vdW formation energies with a per-chemical-system convex hull (13,967 of 19,116 records within 100 meV atom^{-1}). Three of the four JARVIS entries in the original top-30 shortlist fail this recomputed criterion (Mg_6FeBi 0.13, $BaFeHg_4$ 0.36, Ba_4ScHg 0.54 eV atom^{-1}); none of them entered the DFT batch, which is Materials Project only.

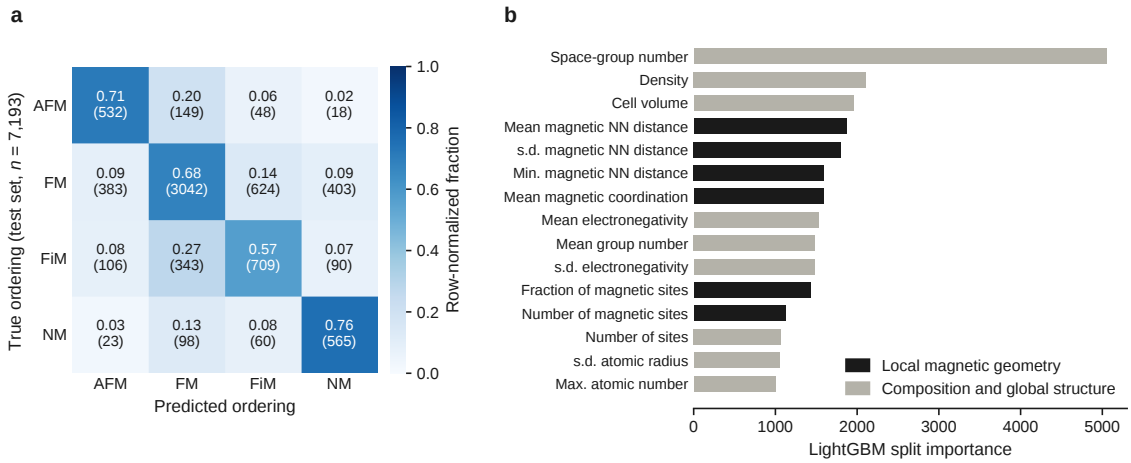
Supplementary Note 3: sampling-window robustness of the Binder crossings

Both crossings were re-derived under changes to the sampling window to test how much of the quoted spread is statistical. The equilibration cut matters and cannot be relaxed: discarding only the first quarter of each run instead of the first half moves mp-754090 to 109.8 K and mp-1079285 to 134.5 K, because the retained data still contain the thermalization transient from the ordered start. With the half-discard in place, mp-754090 is insensitive to further truncation – averaging only the last quarter gives 98.0/102.4/103.0 K, mean 101.1 K, against the published 101.2 K. mp-1079285 is not: its last quarter gives 141.0/141.1/123.4 K, mean 135.2 K, and the entire shift comes from the 2048×4000 pair. A check that shortens no window – analyzing only the six 80 ps runs, which cover 135–150 K – returns 140.0/143.2/135.7 K, mean 139.6 K, within 0.3 K of the published value. The instability of individual pairs is visible directly in the raw series: one 40 ps seed at $N = 4000$, 140 K has $U_4 = 0.582$ in its third quarter and 0.383 in its fourth (a 10 ps window each), so the autocorrelation time of U_4 near T_c is comparable to the run length. Pooling 10–14 seeds suppresses the residual quarter-to-quarter difference of the pooled estimate to ≤ 0.039 but does not eliminate it.

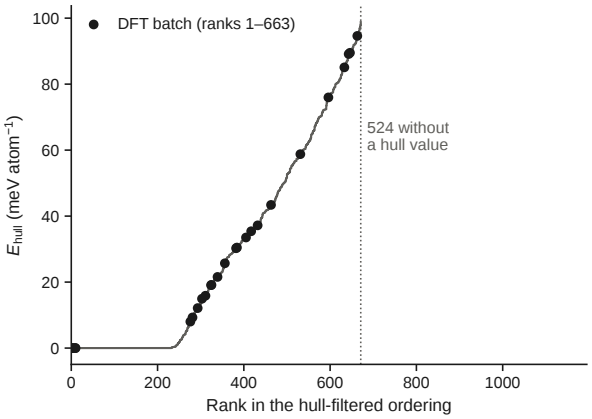
Supplementary Note 4: errors caught by the spin-model checks

The mp-754090 spin model was corrected twice during this work, and the Langevin protocol once; each error was caught by a check that we now apply to every model before a sweep. A first extraction quoted $J = \Delta E/4$, double-counting the flipped bonds ($J = 12.06$ meV, which placed the $|m|(T)$ knee at 150–200 K). The next used a 5.25 Å neighbour cutoff that keeps only the six coplanar in-plane bonds and drops the six inter-layer ones, so that $\Delta E = 8J$ and $J = 5.57$ meV – the correct sum of an in-plane and an inter-layer bond, but on a lattice of uncoupled two-dimensional triangular layers, which has no long-range order at any finite temperature; once properly equilibrated it placed the knee at 25–50 K, and the Monte Carlo on that lattice gives $|m|(10\text{ K}) = 0.64$ and 0.23 at 25 K, the no-long-range-order signature that exposed the cutoff error. Finally, at the conventional damping $\alpha = 0.05$ neither 4 ps nor 20 ps runs had equilibrated at the temperatures nearest the transition (at 100 K the seed-averaged $|m|$ was still falling across the averaged half of both:

from 0.49 in its first 0.5 ps to 0.38 in its last for the 4 ps run, and from 0.15 in its first 2 ps window to 0.07 in its last for the 20 ps run); those runs are kept in the repository and approach the reported values from above.



Supplementary Fig. 1. **Structure-aware ordering classifier on the held-out test set.** n = 7,193 records from a stratified 80/20 split. **a**, Row-normalized confusion matrix (counts in parentheses). The dominant confusion is FM ↔ FiM, between the two net-moment classes that the screening funnel treats identically, whereas AFM and NM are recovered at 71% and 76%. **b**, The fifteen most-used features by LightGBM split count; black, the local-magnetic-geometry features, computed from atomic positions and species only; gray, composition and global-structure features.



Supplementary Fig. 2. **The DFT batch is not the most stable end of the funnel.** Hull distance against rank for the 1,195 candidates that survive the hard hull filter, in an ordering that sorts records with a known hull distance first (671) and by hull distance; the 524 records from sources without a hull value follow the dotted line. Black circles, the 26 DFT candidates, which span ranks 1–663.

Supplementary Table 1. **Ordering classifiers compared on one split.** All models trained on the same records (28,769; the 28,673 with an ordered structure for the graph-based models, of which CGCNN holds out 10% for validation) and scored on the 7,161 test records with an ordered structure (%). CGCNN, mean ± s.d. over three seeds; the LightGBM models are deterministic. MAGN. bal., balanced accuracy on the 369 experimentally labelled test records (311 AFM, 39 FiM, 19 FM); this column varies by up to 4 points with the training set and does not rank the models. MP FM→non-FM, fraction of Materials Project FM test records predicted AFM or FiM by the held-out model, the base rate for the next column. The last two columns (Overt., Conf.) are from retraining with the 26 DFT candidates held out: how many of the 21 overturned candidates are predicted AFM or FiM, and how many of the 3 confirmed ferromagnets are predicted FM (CGCNN, range over seeds). The screening used LightGBM v2 (on all 7,193 test records 67.4% and 68.0%).

		Bal.	Recall				MAGN.	MP FM	Overt.	Conf.
Model	Acc.	acc.	AFM	FM	FiM	NM	bal.	→non-FM	non-FM	FM
LightGBM v2 (screening)	67.4	68.0	70.9	68.4	56.9	75.7	56.0	23.2	10/21	2/3
LightGBM, MACE emb.	69.6	68.3	56.5	71.4	63.9	81.5	53.3	22.0	13/21	2/3
LightGBM, v2 + MACE	74.6	71.7	65.9	78.4	63.0	79.5	55.1	15.8	7/21	2/3
CGCNN	63.3 ± 3.2	65.9 ± 1.0	63.6 ± 6.3	62.0 ± 6.6	58.0 ± 2.3	79.9 ± 4.5	49.1 ± 6.4	32.7 ± 6.7	10–13/21	0–1/3

Supplementary Table 2. **Small-margin overturns re-decided with same-cell pairs.** Margin $E^{\min}_{\text{non-FM}} - E_{\text{FM}}$ (meV atom^{-1}) of the eight candidates whose MP-recipe overturn had $|\text{margin}| < 2$ meV atom^{-1}, before and after the tight same-cell pair, for the configuration in whose cell the pair was built. Every sign is kept. For $Li_4NbFe(WO_6)_2$ another, recipe-only configuration (-0.54) sets the final margin.

Compound	MP id	MP recipe	Same cell
Ta_2FeO_6	mp-1173139	-0.31	-0.31
Ta_2FeO_6	mp-1173143	-0.41	-0.41
$CrFe(BiO_3)_2$	mp-551086	-0.85	-0.80
$FeBi(SeO_3)_3$	mp-1197024	-1.19	-1.19
$CaFeReO_6$	mp-1343817	-1.77	-1.88
Tb_6FeBi_2	mp-22972	-0.14	-0.46
$FeCu_4Re_{14}(Mo_3S_{20})_2$	mp-682554	-0.29	-0.31
$Li_4NbFe(WO_6)_2$	mp-755203	-0.73	-0.47

Supplementary Table 3. **Magnetization of the $Li_4Fe_2TeWO_{12}$ (mp-754090) spin model.** Normalized net magnetization $|m|$ of the 2048-spin Fe sublattice: 20 ps Langevin dynamics ($\alpha = 1$) per point from the ordered state, averaged over the last 10 ps; mean ± s.d. over $n = 3$ thermostat seeds; DFT-derived twelve-neighbour $J = 2.78$ meV, exchange only.

T (K)	$\lvert m\rvert$	T (K)	$\lvert m\rvert$
10	0.966 ± 0.000	150	0.044 ± 0.002
25	0.912 ± 0.001	175	0.039 ± 0.001
50	0.805 ± 0.002	200	0.032 ± 0.002
75	0.659 ± 0.004	250	0.029 ± 0.000
100	0.343 ± 0.027	300	0.027 ± 0.001
125	0.068 ± 0.001	400	0.025 ± 0.000

Supplementary Table 4. **Magnetization of the Tb_6FeBi_2 (mp-1079285) spin model.** As Supplementary Table 3, for the three-shell model ($J_b = 14.61$, $J_{2b} = 3.31$, $J_\perp = 2.79$, $|\mathbf{D}| = 0.098$ meV).

T (K)	$\lvert m\rvert$	T (K)	$\lvert m\rvert$
5	0.988 ± 0.000	60	0.830 ± 0.003
10	0.975 ± 0.000	75	0.778 ± 0.003
20	0.949 ± 0.001	100	0.672 ± 0.003
30	0.921 ± 0.001	125	0.509 ± 0.003
40	0.892 ± 0.002	150	0.128 ± 0.012
50	0.862 ± 0.002	200	0.048 ± 0.002
		250	0.036 ± 0.001
		300	0.033 ± 0.001